

Problem statement:

Does changing only the language, script, or cultural cue change which culturally different but semantically valid images.

Model Selection:

English-centric baseline (CLIP, OpenCLIP), Multilingual models (M-CLIP/XLM-R-based CLIP), Translation/alignment-based multilingual models (NLLB-CLIP), latest VLMs.

Language Selection:

Atleast five Indian languages (of which we should have a decent understanding):

Hindi, Urdu, Bengali, Tamil, Telugu and English (reference)

Prompt design (with different scripts):

English words (like doll), hindi word (like गुड़िया) and Romanized (like gudiya)

Also, does mentioning the context like Indian Doll, भारतीय गुड़िया or Bhartiya gudiya makes a difference.

It will also help us compare implicit cultural association from language with explicit cultural instruction.

Dataset creation:

Dataset should have images of different concepts (Say doll, house, wedding etc). And for each concept, there should be atleast western vs Indian categories. We need atleast 100 images per categories. More is better.

Taxonomy:

Class	Concepts
Class I — Low expected cultural malleability / semantic controls	apple, orange, pear, mushroom, rose, bottle, bee, beetle, butterfly, caterpillar, cockroach, tiger, wolf, camel, chimp, kangaroo, fox, raccoon, lobster, spider, worm, crocodile, lizard, dinosaur, snake, turtle, hamster, rabbit, squirrel, cloud
Class II — Implicitly culturally malleable	doll, toy, house, room, kitchen, bedroom, bathroom, chair, couch, table, lamp, clothes, shirt, pants, shoe, hat, ring, bag, backpack, meal, cafeteria, porch, garden, hotel, school, bus, taxi, train, boat, market

Class	Concepts
everyday concepts	
Class III — Strongly culturally embedded expected cultural responsiveness	architecture, ancient craft, ancient painting, costume, craft, dance, festival, folk tale, folklore character, formal uniform, funeral, headwear, instrument, marriage ceremony, martial art, mask, military parade, mythological figure, native inhabitants, popular street food, pottery style, religious building, ritual, sacred object, social custom, symbolic bird, symbolic plant, tea culture, textile pattern, wedding
Class IV — Socially sensitive / stereotype-prone concepts	doctor, nurse, CEO, homemaker, maid, police officer, scientist, teacher, engineer, politician, judge, criminal, terrorist, thief, gangster, poor, rich, homeless, educated, illiterate, beautiful, attractive, ugly, aggressive, intelligent, lazy, immigrant, tribal, disabled, mentally ill

Positive use-cases (Cultural-responsiveness):

1. If English gives somewhat higher scores to globally/Western-styled dolls while Hindi gives somewhat higher scores to Indian traditional dolls, that need not be harmful.
2. English wedding prefers white gown / church-like wedding imagery while Hindi equivalent suggest Indian wedding attire and ceremony.

Negative use-cases:

1. The term ‘homemaker’ already has biased distribution of gender. Does asking it in Hindi make it more disproportionate?
2. The problem becomes even bigger if terms like “Criminal” give more disproportionate results in certain languages.